Worksheet

For teachers: how to use

Block Programming | National Institute of Technology, Maizuru College — Yasuhito Inoue

Choose and print the sheets that fit your lesson. Sheet 1 is a general sheet for every lesson, sheets 2–7 are for each app, and sheet 8 is a check card for the whole unit.

Sheet	When to use	Grade
1 Goal & reflection	Start and end of every lesson	All grades
2 Drawing	Think about angles (360 ÷ number of corners)	Grade 3+
3 Maze	Draw the path and steps before programming	Grade 2+
4 Arrow maze	Write the steps with arrows	Grades 1–2
5 Space / Break & Build	Track order and counts (collect / carry)	Grade 3+
6 Keys & doors / Slime	Think about state and timing	Grade 4+
7 Python	Write code and keep an error log	Grade 5 – junior high
8 Check card	Record for the whole unit, before the certificate	All grades

- Printing: A4 portrait, one sheet per page.
- Empty boxes are for students to fill in with a pencil. Have students copy the stage board from the app screen or the lesson slides.
- Answers are in the guide (guide.html) and ANSWERS.md. Teacher mode (add ?sensei to the end of the address) loads the answer blocks.

Worksheet

1 Today's goal and reflection

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Class ( ) No. ( ) Name ( ) Date (/)

App I used (circle it)	Stages I did
Drawing / Maze / Arrow / Keys & Doors / Slime / Space / Break & Build / Python	Stage to

Today's goal (write in your own words)

Goal

Reflection (circle one)

Question	Great	Good	Almost
Did I reach my goal?	◎	○	△
Did I plan the steps before running?	◎	○	△
When it didn't work, did I fix it and try again?	◎	○	△
Did I help and learn from friends?	◎	○	△

What I learned / what was fun today

What was hard / what I want to try next

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2 Block Drawing: think about angles

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Name ( )

When the turtle turns all the way around, it turns 360 degrees. So the turn angle = 360 ÷ number of corners. Fill in the blanks.

Shape	Corners	Repeat times	Turn angle (°)	Calculation
Triangle	3	3		360 ÷ 3 =
Square	4			360 ÷ =
Hexagon		6		
Octagon	8			
Circle (36 small steps)		36		
Star (turns around twice)	5	5		720 ÷ 5 =

Try it: draw a shape you want to make, and write the blocks you will use.

My shape (drawing)	Blocks to use (in order)
	1.
	2.
	3.
	4.
	5.

What I noticed when I ran it (if it was different from the sample, what did I fix?)

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3 Block Maze: draw the path and think

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Name (　　)　Stage (　　)

1 Copy the maze

Start = S, Goal = G, rock = ■. Draw the explorer's path with a pencil line.

	1	2	3	4	5
1					
2					
3					
4					
5					

2 Write the steps

use "forward", "right" and "left" in order. Pretend you are the explorer to decide right and left.

No.	Step	No.	Step
1		7	
2		8	
3		9	
4		10	
5		11	
6		12	

3 Make it shorter

Question	Answer
Where does the same move repeat?	
With "repeat ○ times", how many blocks do you need?	blocks
What gets better with "if front is open"?	

Result (where did it stop? how did you fix it?)

Worksheet

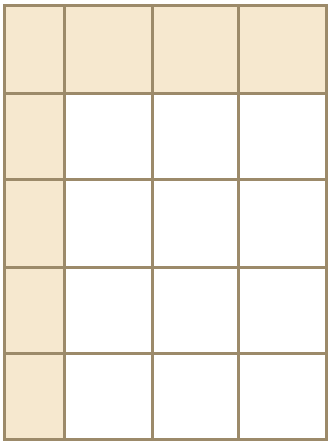
4 Arrow Maze: write the arrows

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Name (　　)　Stage (　　)

1 Draw the maze

mark the start ○, the diamond ☆ and the bushes ×.



2 Write the arrows in order

(↑ ↓ ← →)

1	2	3	4	5	6	7	8

3 Count

Arrow	How many?
↑ up	blocks
↓ down	blocks
← left	blocks
→ right	blocks
Total	blocks

4 Did you do it?

Color a ☆ when you reach the diamond.　☆　☆　☆

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5 Space / Break & Build: track the order and the counts

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Name ( ) App (Space / Break & Build) Stage ( )

After each block runs, write how the number of "stars collected" or "blocks carried" changes.

No.	Action (forward, collect, break, place…)	Stars or blocks
Start	—	0
1		
2		
3		
4		
5		
6		
7		
8		

Question	Answer
What happens if you "collect" where there is no star?	
What happens if you "place" with no blocks?	
Why is it good to check with "if …" before moving?	

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6 Keys & Doors / Dodge the Slime: state and timing

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Name ( )

🗝️ Keys & Doors Stage ( ): write how the number of keys changes.

No.	What I did	Keys held
Start	—	0
1		
2		
3		
4		
5		

Question	Answer
What must you do before opening a door?	
What happens if you walk into a door with no key?	

👾 Dodge the Slime Stage ( ): predict where the slime will be.

Turn	Explorer (forward, wait…)	Where is the slime?	Danger? (○ / ×)
1			
2			
3			
4			
5			
6			

When is it good to use "wait"?

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7 Python: write code and keep an error log

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Name (　　)　App (　　)　Stage (　)

1 Blocks → Python

rewrite the blocks in Python.

Block	Python
move forward	
turn right	
repeat 4 times	
repeat until goal	
if front is open	
else	

2 Write the code

you may write the indent (4 spaces) as "→".

Line	Code
1	
2	
3	
4	
5	
6	
7	
8	

3 Error log

an error is not a failure, it is a hint! Write down the errors you get.

Line	Error name (SyntaxError etc.)	What was wrong?	How did I fix it?

Common errors: missing ":" (SyntaxError), wrong indent (IndentationError), missing "()", and spelling mistakes (NameError).

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8 Check card

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Name ( )

Put a ✓ on each stage you cleared (for Drawing, each task). "—" means there is no stage.

App	1	2	3	4	5	6	7	8	9
🐢 Drawing									
⛏️ Maze									—
🧭 Arrow Maze							—	—	—
🗝️ Keys & Doors					—	—	—	—	—
👾 Dodge the Slime					—	—	—	—	—
🚀 Space								—	—
🧱 Break & Build								—	—
🐍 Python Drawing									
🐍 Python Maze						—	—	—	—
🐍 Python Space							—	—	—
🐍 Python Break & Build								—	—
🐍 Python Arrow							—	—	—
🐍 Python Keys & Doors					—	—	—	—	—
🐍 Python Slime					—	—	—	—	—

When you clear every stage of an app, get your certificate from "⚙ Settings" → "📜 Certificate".

My favorite app, and why